

FHIR R4 patient records on Databricks

Manager report · 30 September 2026 · Author: Usama Gul · Platform: Databricks

The data is synthetic. It was generated by Synthea (MITRE) and published free by Databricks: realistic patient histories, but no real people. Every number below describes the pipeline and this synthetic population — none of it is a real-world health statistic.

1,156

FHIR files loaded (2.08 GB),
in 5 batches

631,630

records, equal to an
independent profile of the
source

24 of 24

record types match that
profile exactly

0 / 0 / 0

records lost · duplicate ids ·
quarantined issues

1. Summary

- What:** an automated Databricks pipeline that turns FHIR R4 — the world standard for exchanging medical records, as deeply nested JSON — into clean, linked, analysis-ready tables.
- Proven complete:** all 1,156 source files and **631,630 records** loaded; the counts of all 24 record types equal an independent profile of the source, measured before building.
- Proven incremental:** loaded in 5 batches; each run processed only the new files (~2 minutes per batch, whatever the total).
- Quality enforced:** automatic gates stop a run on lost records, duplicate ids, broken files or failed rules; problems are quarantined with a reason, never silently dropped. Final count: 0.
- Delivered:** a 7-page, 55-chart Power BI dashboard, generated from code and kept in git.
- Not yet:** production deployment, one automation Job, 13 smaller record types (section 7).

2. The dataset

Item	Measured
Source	Databricks Marketplace, "FHIR R4 Synthetic Data" (free). Its notebook "FHIR R4 Read with SQL" reads a public S3 folder.
Files	1,156 = 1,154 patient bundles + 2 reference bundles (hospitals, practitioners); 2.08 GB
Updates	None. All files dated 2022-04-15: a static snapshot. Batches were used to practise real incremental loading.
Population	1,154 patients (1,000 alive, 154 deceased), all in Massachusetts; visits 1920–2022
Why Databricks only	The Snowflake Marketplace has FHIR data from a different maker — the same standard but different data, so no fair platform comparison was possible.
The Marketplace share	Contains no tables (only the notebook). The pipeline reads the raw files, as the notebook does.

3. What was built

Landing	Copy job moves files from the public bucket into our own volume, in batches; never overwrites.	1,156 files
▼ Auto Loader: only new files		
Bronze	One row per file: the raw text (evidence) + the parsed JSON (VARIANT type, any shape).	1,156 rows
▼		
Silver	Every record unpacked, then 11 typed tables — one per record type, each with its own columns — plus quarantine.	631,630 records
▼		
Gold	11 chart-ready tables: age bands, visit classes, clinical vs social conditions, 18 vital-sign measures, costs.	11 tables
▼		
Gate	Stops the run if anything is lost, duplicated or broken. Nothing half-built is published.	6 rules
▼		
Dashboard	Power BI Project generated by two scripts: 7 pages, 55 visuals.	in git

Everything is defined as code (Databricks asset bundle with dev and prod targets) and kept in a git repository. No secrets in code.

4. The evidence

4.1 Complete and correct

Check	Expected	Pipeline	
Files loaded	1,156	1,156	Match
Records in = records out (bronze → silver)	631,630	631,630	Match
Record types with identical counts	24	24	Match
Patients: alive / deceased	1,000 / 154	1,000 / 154	Match
Visits = clinical notes = insurer statements	46,050 each	46,050 each	Match
Records lost when flattening into typed tables	0	0	Pass
Duplicate record ids	0	0	Pass
Quarantined issues	—	0	Pass

"Expected" comes from profiling the source files *before* building — on a laptop for a 50-file sample, and inside Databricks for all files. Two independent measurements agreeing is the proof.

4.2 Incremental loading — pipeline run time per batch

Batch	Files	MB	Pipeline	0 – 131 s
1	52	86.2	52 s	
2	276	598.6	131 s	
3	276	452.5	114 s	
4	276	425.4	117 s	
5	276	515.7	107 s	

Run time follows the size of the *new* batch, not the total already loaded: each file is loaded exactly once. Single runs on serverless compute; indicative, not a benchmark.

4.3 Cost

Databricks cost is **NOT MEASURED**: billing data cannot be read on this trial account, and no spend ceiling is available on it. Everything runs serverless and stops when idle; each run was approved with an estimate beforehand.

5. What the data shows (synthetic)

Records	Count	Highlights	Value
Visits	46,050	Outpatient share of visits	42,788 of 46,050
Observations (vitals, labs, surveys)	162,313	Claims total (synthetic prices)	\$155,580,266
Procedures	61,275	Average BMI / systolic BP / HbA1c	26.3 / 122.9 / 5.5%
Prescriptions	44,808	"Conditions" that are social findings	25,697 of 32,076
Conditions	32,076	Most prescribed	lisinopril (BP)
Vaccines	9,577		
Claims	90,858		

The dashboard has seven pages: Overview, Visits, Diagnoses, Treatments, Vitals & labs, Costs, Data quality. One insight it makes visible: 80% of recorded "conditions" are social findings (employment, stress, isolation), so the dashboard separates them from clinical diseases.

6. Findings worth knowing

- F1 — Checks caught wrong assumptions before wrong numbers were published.** A rule "every location has an identifier" stopped a run. The record was "Patient's Home" — a FHIR *kind* of place, not a specific address. The rule was corrected, and home visits (123 visits, 13 procedures) are now matched and labelled.
- F2 — The same field has different shapes in different record types** (an address is a list for patients, a single object for locations). Reading everything through one schema breaks; the pipeline gives each record type its own table and keeps the raw JSON in a flexible type.
- F3 — Serverless jobs cannot reach the open internet.** The file copy was switched to Databricks' built-in storage connector.

F4 — Auto Loader remembers file names, not contents. A renamed file would load twice; the gate now stops the run on duplicate ids, and the landing folder is treated as write-once.

7. Not done yet, and next steps

Item	Status
Production deployment	Next Built and verified in the development environment; the production target is defined but not deployed.
One automation Job (copy → pipeline → checks) with a trigger	Next Each step exists and was run; they are not yet chained and scheduled.
13 smaller record types (claims detail, reports, care plans...)	Next Loaded and kept whole; not yet split into their own tables.
110 location links inside those types	NOT MEASURED — answered when the types above are typed.
Runbook and error log (as for the BRFSS project)	Next
Measured cost	NOT MEASURED — needs an account where billing is readable.

Evidence: `docs/report/scope_and_gaps.md` (decisions and results D1–D13), `docs/design/dataset_profile.md` (profile), `docs/design/design.md` (design), `docs/learning/walkthrough.md` (full explanation). Data: Synthea synthetic patients via the Databricks Marketplace.